Congruence separation and positive-word fusion

Smaller explicit Diophantine consistency equations for ProveIt

AI-assisted research artifact prepared for the ProveIt project

Mathematical constructions, literal circuits, and independent finite audits

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Abstract

We construct explicit integer polynomials whose unsolvability is equivalent in ZFC to the syntactic consistency of ZFC, by improving the arithmetic reductions attached to ProveIt's fixed refutation-search compiler. The main construction has ten unrestricted integer variables, exact total degree 9,152,284, and a division-free circuit of 586 binary arithmetic operations. The strongest incoming construction at the pinned repository snapshot has degree 21,111,400 and 659 operations in the same domain. The principal new step combines two square conditions and a positivity condition in one coprime-factor guard. Congruences distinguish that radicand from the remaining Lucas radicand, so a four-factor norm suffices. For nine positive variables we obtain degree 5,671,669 with 630 operations; a cheaper variant has degree 6,692,108 with 604 operations. A secondary fourteen-positive-variable construction has degree 52 and 79 operations. A common source transformation reduces 136 weighted records to 94, with a newly checked selective code. The archive includes all literal circuits, coefficient payloads and bounded recipes, proofs, generators, and independent finite checks. The reductions are ordinary mathematics formalizable in ZFC; no new end-to-end proof-assistant certificate or global minimality claim is asserted.

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1 Results, objective, and logical meaning

There is no unique mathematical ordering called "simplest Diophantine equation." Number of supplied variables, their domain, total degree, straight-line evaluation cost, coefficient height, and proof complexity are different criteria. We retain this distinction and improve concrete points of the supplied repository's frontier. We do not claim a lower bound or an optimal equation among all integer polynomials.

Theorem 1.1 (Main constructions)**.** *Let* Con(ZFC) *mean that no finite proof in the repository's fixed recursive presentation of ZFC derives a contradiction. The literal circuits specified in this report and the accompanying data define fixed polynomials* $F_{10} \in \mathbb{Z}[\mathbf{t}]$*,* $F_9, F_{9,\mathrm{f}} \in \mathbb{Z}[\mathbf{x}]$*, and* $P_{14} \in \mathbb{Z}[\mathbf{u}]$*, with respectively ten, nine, nine, and fourteen variables. Composing the proved arithmetic reductions below with the inherited calibrated source theorem gives*

$$\mathrm{ZFC} \vdash (\forall \mathbf{t} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{t}) \neq 0) \iff \mathrm{Con(ZFC)},$$
$$\mathrm{ZFC} \vdash (\forall \mathbf{x} \in \mathbb{Z}_{>0}^{9}\ F_9(\mathbf{x}) \neq 0) \iff \mathrm{Con(ZFC)},$$

and the same statement holds for $F_{9,\mathrm{f}}$ *and for* P_{14} *over* $\mathbb{Z}_{>0}^{14}$*. Their exact degrees and emitted circuit costs are those in Table 1.*

The phrase "inherited calibrated source theorem" is substantive. It is the fixed positive-source-to-blank-halting equivalence and the calibration of the literal K215 machine to syntactic ZFC refutations, not an assumption that ZFC is consistent. Section 7 identifies the proof dependencies and the separate formal-verification boundary. Each new arithmetic reduction can also be read as an unconditional theorem about existence of zeros of the explicitly retained positive source.

Construction and supplied domain	Variables	Exact degree	Operations
New two-root integer construction	10 in $\mathbb{Z}$	9,152,284	586
Incoming selective integer baseline	10 in $\mathbb{Z}$	21,111,400	659
New majorant construction	9 in $\mathbb{Z}_{>0}$	5,671,669	630
New cheaper majorant variant	9 in $\mathbb{Z}_{>0}$	6,692,108	604
Incoming selective positive baseline	9 in $\mathbb{Z}_{>0}$	7,344,328	738
New auxiliary-gap construction	14 in $\mathbb{Z}_{>0}$	52	79
Incoming support-sensitive baseline	14 in $\mathbb{Z}_{>0}$	56	79

Table 1: Like-for-like comparisons at commit `4b4349064a75`. Every operation count is for a particular emitted live circuit, not a minimal-circuit theorem.

The ten-integer degree decreases by approximately 56.65%, with approximately 11.08% fewer operations. The nine-positive majorant degree decreases by approximately 22.77%, with approximately 14.63% fewer operations. The number of variables is unchanged in each comparison. The nine-variable results also have natural-coordinate versions: replacing each positive coordinate by a natural coordinate plus one preserves the degree. The emitted natural circuits use 636 and 610 operations.

These comparisons are against the actual incoming archives, including the latest selective package, rather than only the older repository summary. Other existing points remain useful: the incoming sparse-Pell package has a nine-positive circuit of degree 15,502,748 and 548 operations, and a ten-integer circuit of degree 50,822,936 and 582 operations. The incoming short constructions include fifteen-positive circuits of degree 44 with 75 operations and degree 157 with 66 operations. Thus the present work does not establish the fewest-operation record over all variable counts.

An operation is a live binary addition, subtraction, or multiplication; squaring and multiplication by a numeral are charged. Shared gates are charged once. Literal and fixed-coefficient leaves

are not charged. Fixed coefficient generation and the bit cost of enormous integers are reported separately. No variable exponentiation or division occurs in any final circuit. The full monomial expansions would be unhelpfully large; a finite acyclic arithmetic circuit is an exact definition of an integer polynomial.

The theorem proves an equivalence, not either of its two sides. In particular it is not a proof of Con(ZFC) or of unsolvability. No consistency assumption is used to select the coefficient tuple. This avoids both a nonconstructive "choose zero or one depending on consistency" definition and any conflict with the second incompleteness theorem.

2 The fixed source and positive-word fusion

2.1 The retained positive quadratic system

Let $p_1, \dots, p_{22}$ be the original package-31 residuals in 31 positive coordinates, listed in Appendix A. Their four fixed ports are (β, K_0, M_C, M_F), with M_F already shifted as required by that source. We use the whole `wang943-mixed-121x8801` tuple, never a mixture of individual constants from different compilers. Its native radix is $2^{1064921}$. The fixed integer bit lengths are

$$(1064921,\ 1070004,\ 1064921,\ 1064922).$$

The four unsigned little-endian payloads contain 533,099 bytes in total before compression. The archive contains the payloads, their byte hashes, their finite compiler recipe, and the literal machine and tile data from which they are generated. In particular these are instantiated integers, not four extra quantified variables.

The inherited source theorem is

$$\exists \mathbf{z} \in \mathbb{Z}_{>0}^{31} \bigwedge_{j=1}^{22} p_j(\mathbf{z}) = 0 \quad \longleftrightarrow \quad \exists n\ \mathrm{Refutation}_{\mathrm{ZFC}}(n). \tag{1}$$

The unbounded native-window compiler and calibrated-machine proof are retained dependencies. The new source arithmetic below proves equivalence to the left-hand side of (1); it does not replace that compiler with a finite simulation.

2.2 A global shift of the packed word

Write $p_j(\mathbf{z}) = \sum_{|\alpha| \le 2} c_{j,\alpha} \mathbf{z}^\alpha$ and set

$$d_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, c_{j,\alpha}.$$

Choose the weights w_i and row targets q_j of Appendix B, and define the fixed sparse polynomial

$$\mathcal{D}(T) = \sum_{j,\alpha} d_{j,\alpha} T^{q_j - \sum_i w_i \alpha_i}, \qquad \ell(T) = \sum_{i=1}^{31} T^{w_i}, \quad \mathcal{A}(T) = \sum_{j=1}^{22} T^{q_j}. \tag{2}$$

All exponents in $\mathcal{D}$ are nonnegative. The largest coordinate weight is $W = 183$, the largest target is $Q_c = 5190$, and

$$K_c = Q_c + 2W = 5556, \qquad E_c = K_c + W + 2 = 5741.$$

The certificate enumerates each of 94 supported source records against all $\binom{33}{2} = 528$ padded input pairs. Exactly 94 of the 49,632 interactions hit marked targets, and every hit belongs to its

intended row and monomial. Padding uses coordinate zero of weight zero. The factorial factor above times the expansion multiplicity is always two. Consequently

$$[T^{q_j}]\left(1+\sum_i z_i T^{w_i}\right)^2 \mathcal{D}(T) = 2p_j(\mathbf{z}). \tag{3}$$

This is an exact polynomial identity, not an empirical statement about a sample of large integers.

Previously each residual was expanded at $\mathbf{n}+\mathbf{1}$, producing 136 weighted records. We instead encode the nonnegative offsets by $g=\sum_i n_i B^{w_i}$ and use

$$1+g+\ell(B) = 1+\sum_i (n_i+1)B^{w_i}. \tag{4}$$

The positive source can therefore remain unexpanded, with only 94 records. This is a substitution in the common packing word rather than 22 separate translations. There is an essential finite correction: translation had canceled two monomials, so the old shifted-support code did not certify the original support. We retarget row 13 from 2212 to 2124 and row 21 from 1050 to 1059. All other weights and targets remain unchanged. The legacy code is retained as a negative control: it has five forbidden interactions on the new support. The corrected code passes the complete enumeration.

2.3 Bounds and the sparse borrow argument

Put

$$M = 2^{2134927}, \qquad \kappa = 2134940, \qquad a_c = 2^\kappa = 8192M, \qquad B = a_c b^2.$$

The actual coefficient recipes give $|[T^r]\mathcal{D}(T)| < M$. Their largest triangle bound has bit length 2,134,926. Independently checked support data give $\deg \mathcal{D} = 5102$ and leading coefficient one. Thus for $b \geq 2$,

$$\mathcal{D}(B) > B^{5102} - M\frac{B^{5102}-1}{B-1} > 0.$$

The following common registers will be used by both few-variable routes:

$$\begin{aligned} N_0 &= B^{184}, & Z_c &= B^{5557}, \\ M_c &= N_0 - 1 - (b-1)\ell(B), & T_c &= M_c + N_0(B-2)\mathcal{A}(B), \\ J_c &= 2(1+g+\ell(B))^2\mathcal{D}(B) + \mathcal{A}(B), & S_c &= g + N_0 J_c. \end{aligned} \tag{5}$$

Under the larger of the two needed bounds, $0 < g < 4bB^{183}$, the distinct positive weights imply $1+\ell(B) < bB^{183}$ and $1+g+\ell(B) < 5bB^{183}$. Hence

$$0 < \frac{J_c}{Z_c} < \frac{50Mb^2+1}{B-1} < \frac{1}{128}.$$

The last inequality reduces to $129 < 1792Mb^2$. Also $0 < T_c < N_0 Z_c/128$ after the same elementary geometric estimates, and $S_c < N_0 Z_c/64$; the margins are more than sufficient for either block length used below. These bounds precede digit decoding.

Suppose now that b is a power of two. The low mask condition $g \,\&\, M_c = 0$, with $g < N_0$, says exactly that

$$g = \sum_i n_i B^{w_i}, \qquad 0 \leq n_i < b.$$

The positive coordinates $z_i = n_i + 1$ lie in $[1, b]$. Coefficients of $(1+\sum z_i T^{w_i})^2\mathcal{D}(T)$ have absolute value less than

$$M(1+31b)^2 < 1024Mb^2 = B/8.$$

Raw coefficients of J_c are therefore bounded in absolute value by $B/2$. At target q_j their exact value is $4p_j(\mathbf{z}) + 1$. Base-B normalization has incoming carries only in $\{-1, 0\}$: induction uses $\lfloor (c+\varepsilon)/B \rfloor \in \{-1, 0\}$ in this range. The marked mask $B-2$ permits precisely digits zero or one. Such a digit cannot have outgoing carry -1, since the required raw value would then be near $-B$, outside the bound. The target digit is thus $4p_j(\mathbf{z}) + 1 + \varepsilon$, with $\varepsilon \in \{-1, 0\}$. It is zero or one exactly when $p_j(\mathbf{z}) = 0$. Positivity of J_c ensures termination of the borrows. No infinite negative radix expansion is being treated as a nonnegative word.

Conversely, given a positive source zero, choose an arbitrarily large power of two b and encode $n_i = z_i - 1$. The word g is positive: the all-one source vector is not a zero, because residual 5 is one there, independently of all fixed ports. It satisfies $g < bB^{183}$ and is divisible by B, hence by b^2. These facts will justify the congruence restrictions used later. They are properties of the constructive source witnesses, not assumptions about arbitrary solutions of the final equations.

3 Two algebraic tools

Lemma 3.1 (Unweighted two-root norm)**.** *Let $a, d \in \mathbb{Z}$ with $a \neq d$. If $t \in \mathbb{Q}$ satisfies*

$$(t^2 - a - d)^2 - 4ad = 0,$$

then a, d are integer squares and t is an integer. Conversely any signed integer roots $u^2 = a, v^2 = d$ give a zero with $t = u + v$.

Proof. At $t = 0$ the left side is $(a-d)^2$, so $t \neq 0$. Define

$$u = \frac{t^2 + a - d}{2t}, \qquad v = t - u.$$

Direct expansion gives $u^2 = a$ and $v^2 = d$. A rational number whose square is an integer is an integer, by writing it in lowest terms. Thus u, v, t are integers. The converse is the same identity. □

There is no premise about the sign of the candidate radicands and no square-class independence premise. With exactly two radicals, an unconditional inequality $a \neq d$ is enough. Replacing large numerical separators by cheap congruences is the central design change.

Lemma 3.2 (Coprime square and positivity guard)**.** *Suppose $b, Q \in \mathbb{Z}$, $|b| \geq 2$, $\gcd(b, Q) = 1$, and $b \equiv Q \equiv 1 \pmod 3$. Suppose also $Q \neq 0$, and set $r = bQ$. For an integer O define*

$$A_O(z) = r + (3O - 1)(rz)^2.$$

There is a nonzero integer z making $A_O(z)$ a square if and only if b, Q are positive squares and $O > 0$.

Proof. Factor $A_O(z) = r[1 + (3O-1)rz^2]$. The factors are coprime and nonzero: the second is one modulo r, and $|r| \geq 2$. If their product is a square, they are signed squares with the same sign. Since $r \equiv 1 \pmod 3$, it cannot be a negative square. Both factors are positive squares. Applying the same coprime-factor argument to $r = bQ$, and again using residue one modulo three, shows that b, Q are positive squares. If $O \leq 0$, the second factor is at most $1 - r < 0$, a contradiction.

Conversely, r is a positive square, and $3O - 1$ is a positive nonsquare, being two modulo three. Thus $(3O-1)r$ is a positive nonsquare. Pell's theorem supplies $u, z > 0$ with $u^2 - (3O-1)rz^2 = 1$. Then $A_O(z) = ru^2$ is a square. □

Lemma 3.3 (The retained quintic merger)**.** *The polynomial* $\pi(T) = (T^3 - T - 1)(T^2 + 23)$ *has no rational root and has a root modulo every positive integer. For* $S \neq 0$*, put*

$$\Theta = (\rho^3 - \rho S^2 - S^3)(\rho^2 + 23S^2), \qquad \rho = T + Sy.$$

Then $\Theta \neq 0$*. If* $\xi S^5 \mid \Theta$*, then* $\xi \neq 0$ *and* $S \mid T$*. When* $S \mid T$ *and* $\xi \neq 0$*, the divisibility can be arranged by a prescribed residue class of* y *modulo* ξ*.*

Proof. The rational-root theorem excludes a root of the cubic; the quadratic is positive over $\mathbb{Q}$. For an odd prime $p \neq 23$, either the cubic has a simple root or it is irreducible. In the second case its Frobenius permutation is a three-cycle, so its discriminant -23 is a square modulo p; the quadratic then has a simple root. At $p = 23$ the cubic has the simple root 3. Hensel lifting handles odd prime powers. For powers of two, $-23 \equiv 1 \pmod 8$ is an odd square in $\mathbb{Z}_2$; the elementary successive lifting of odd squares gives the same conclusion. The Chinese remainder theorem combines the roots.

Now $\Theta = S^5\pi(\rho/S)$ cannot vanish. If a prime satisfies $\beta = v_p(T) < v_p(S) = \alpha$, then $v_p(\rho) = \beta$. The two factors of Θ have valuations exactly 3β and 2β, so $v_p(\Theta) = 5\beta < 5\alpha$, contradicting $S^5 \mid \Theta$. Finally, when $T = Sq$, choose c with $\xi \mid \pi(c)$ and impose $y \equiv c - q \pmod \xi$. □

The quintic merger was already available in the supplied constructions; it is retained here, not claimed as a new invention.

4 The ten-integer equation

4.1 All polynomial definitions

The ten supplied variables are

$$(t, g_0, h_0, k_0, l, w, x, y, z, m) \in \mathbb{Z}^{10}.$$

The circuit names the first coordinate `f`. Define

$$b = 6t + 4, \quad g = bg_0, \quad h = bh_0, \quad k = b(3k_0 + 2), \quad B = a_c b^2. \tag{6}$$

In the inherited bridge the old first coordinate is $f = 2t + 1$, so $b = 1 + 3f$. The emitted circuit evaluates $6t + 4$ directly. Use S_c, T_c from (5), and set

$$\begin{aligned} N_c &= B^{5741}, & X &= N_c^2 + (S_c + T_c)N_c + N_c - 1 - S_c, \\ Y &= a_c^{15}\, 2^{2134919}\, B^{5931} b, & \mathscr{C} &= bB^{183}. \end{aligned} \tag{7}$$

The following registers use the modified Lucas bridge of Sun [5, 6]:

$$\begin{aligned} &L = lY, && U = 2LX, && V = 4gwY, && W_* = bw, \\ &A = U(V + 1), && B_* = 2X + 1, && C = B_* + (A - 2)h, \\ &P = U^2V, && K = X + 1 + k(P - 2), && S = 2A - 5, \\ &T = 3W_*C - 2(W_*^2 - 1), && Q = (P_{1/2}^2 - 1)K^2 + 1, \end{aligned} \tag{8}$$

where $P_{1/2} = (LX)UV$ is a polynomial and $2P_{1/2} = P$. In particular $4Q = (P^2 - 4)K^2 + 4$, the bridge's auxiliary square expression. Put

$$\Delta = 8\mathscr{C}^3 gK^2 - g^2\left(32(C - KL)^2\mathscr{C}^3 + g^2K^2\right). \tag{9}$$

For $\xi = zx$ define

$$\begin{aligned} D &= (A^2-4)C^2+4, & E &= C^2D\xi, \\ F &= 4(A^2-4)E^2+1, & G_* &= 2(A+2)(A-2)^2E^2-1, \\ H_* &= C+(B_*-C)F+16EFy, & I_* &= (G_*^2-1)H_*^2+1. \end{aligned} \tag{10}$$

Finally let

$$\begin{aligned} \rho &= T+Sy, & \Sigma &= \xi S^5, & \Theta &= (\rho^3-\rho S^2-S^3)(\rho^2+23S^2), \\ r &= bQ, & a &= r+(3\Delta-1)(rz)^2, & d &= DFI_*, \\ Z &= \Sigma m-\Theta. \end{aligned} \tag{11}$$

The requested single equation is

$$\boxed{F_{10} = \Big(Z^2-\Sigma^2(a+d)\Big)^2 - 4\Sigma^4 ad = 0.} \tag{12}$$

Every letter on the right is a computed abbreviation. Only the ten listed coordinates are unknowns. The equations defining the abbreviations do not add existential witnesses.

4.2 The exact imported interface

For clarity, the number-theoretic input used here is stronger than a bare existence theorem for some ten-variable representation. Sun's 2021 Theorems 4.12–4.13, specialized at $p = P_{\mathrm{thm}} = 2$ and $Q_{\mathrm{thm}} = 1$, and the modified certificate and interpolation of the 2026 preprint give the following interface.

From a valid source encoding with b a power of two, $0 < g < \mathscr{C}$, and $Y \mid \binom{2X}{X}$, the constructive proof chooses positive l, w and integral h, k such that

$$C = u_{2X+1}(A), \quad K = u_{X+1}(P), \quad Q \in \mathrm{Sq}(\mathbb{Z}), \quad S \mid T, \quad \Delta > 0.$$

Here $u_0(A) = 0, u_1(A) = 1, u_{n+1}(A) = Au_n(A) - u_{n-1}(A)$. The exact C and K choices occur immediately before and in equation (4.16) of the 2021 proof. They matter for the new residue restrictions. The interpolation theorem allows ξ to be divisible by any prescribed positive integer and subsequently allows any residue class of y modulo ξ, while preserving $DFI_* \in \mathrm{Sq}(\mathbb{Z})$.

Conversely, for the displayed registers the hypotheses

$$b \in \mathrm{Sq}(\mathbb{Z}), \quad f^2l^2\Delta > 0, \quad \xi \neq 0, \quad 4Q \in \mathrm{Sq}(\mathbb{Z}), \quad DFI_* \in \mathrm{Sq}(\mathbb{Z}), \quad S \mid T$$

recover b as a power of two, $0 < g < 2\mathscr{C}$, and the binomial divisibility. This is the interface of the retained reverse bridge; its source size conditions hold after the positive-word change.

Indeed, for $b = 2^s$ and $B = 2^v$, the explicit mask has

$$\mathrm{pop}(T_c) = 206v - 31s - 22.$$

The two-block definition of X, with $0 \le S_c, T_c$ and $S_c + T_c < N_c$, gives

$$\mathrm{pop}(X) = 1 + 5741v + \mathrm{pop}(T_c) - \{\mathrm{pop}(S_c) + \mathrm{pop}(T_c) - \mathrm{pop}(S_c + T_c)\}.$$

Since $v_2\binom{2X}{X} = \mathrm{pop}(X)$ and $\log_2 Y = 5947v - 31s - 21$, divisibility by Y is exactly absence of the indicated binary carries. Section 2 then decodes the source zero. For the forward implication we may take any sufficiently large $b = 4^s$, so it is also a square. This explains why the more specialized source still meets the imported interface.

4.3 Universal congruences and radicand separation

The following statements hold on every signed assignment, before assuming the final equation or any positivity. We have $b \neq 0$, $|b| \geq 2$, $2 \mid b$, $b \equiv 1 \pmod 3$, and $a_c \equiv 1 \pmod 3$. Both N_0 and N_c are divisible by b^2, so

$$X + 1 \equiv -g \equiv 0 \pmod b.$$

Also X is odd. Modulo three, all powers of B are one, $M_c = 0$, and $T_c \equiv -22 \equiv 2$. The terms involving S_c cancel, giving $X \equiv 1 + T_c \equiv 0 \pmod 3$.

The factor b in Y gives $2b \mid U$ and $2b \mid A$. Hence $b \mid P$, and $3 \mid P$ follows from $3 \mid X$. The imposed k gives

$$b \mid K, \qquad 3 \mid K, \qquad Q \equiv 1 \pmod{b^2}, \qquad Q \equiv 1 \pmod 3.$$

Thus $Q \neq 0$, Q is odd, and $\gcd(b, Q) = 1$. Also

$$C + 1 = 2(X + 1) + (A - 2)bh_0 \equiv 0 \pmod{2b}.$$

Writing $D = A^2C^2 - 4(C^2 - 1)$ shows $4b \mid D$. Therefore

$$\frac{a}{b} = Q + (3\Delta - 1)bQ^2z^2 \equiv 1 \pmod 2, \qquad \frac{d}{b} \equiv 0 \pmod 2. \tag{13}$$

The quotients in this proof are integers; they are not circuit operations. As $b \neq 0$, equation (13) proves $a \neq d$ unconditionally. A large separating weight is unnecessary.

4.4 Soundness, including all degenerate cases

Suppose $F_{10} = 0$. The integer $S = 2A - 5$ is odd and nonzero. Lemma 3.3 gives $\Theta \neq 0$. If $\Sigma = 0$, the final polynomial would equal Θ^4, a contradiction. Thus $\Sigma \neq 0$, in particular $\xi \neq 0$ and $z \neq 0$.

Divide the norm identity by Σ^4 and apply Lemma 3.1 with $t = Z/\Sigma$. It yields $a, d \in \mathrm{Sq}(\mathbb{Z})$ and $Z/\Sigma \in \mathbb{Z}$, hence $\Sigma \mid \Theta$. The quintic merger gives $S \mid T$. Lemma 3.2 now gives b, Q positive squares and $\Delta > 0$.

As $\mathscr{C} > 0$, formula (9) forces $g > 0$ and $K \neq 0$: if $g \leq 0$ every contribution is nonpositive, and if $K = 0$ the remaining term is nonpositive. If $l = 0$, then $U = P = P_{1/2} = 0$ and $Q = 1 - K^2 \leq 0$, again impossible. Thus $l \neq 0$. The old $f = 2t + 1$ never vanishes, so $f^2l^2\Delta > 0$. This recovers the inherited nonvanishing hypothesis although its f^2l^2 factor has been omitted from the new guard. Since Q is a square, $4Q$ is a square. All hypotheses of the reverse interface now hold, and it decodes a positive source zero. No prior sign assumption on d or on the free certificate variables was used.

4.5 Completeness of the normalization

Suppose the positive source has a zero. Choose sufficiently large $b = 4^s$ with $s \geq 1$. Then $t = (b-4)/6$ is an integer, and its canonical packed g is positive and divisible by b^2. Hence $g = bg_0$, and $b^2 \mid X + 1$. Use the constructive bridge, retaining its exact $C = u_{2X+1}(A)$ and $K = u_{X+1}(P)$.

An even-index Lucas polynomial is divisible by its parameter. As $X + 1$ is even and $4b^2 \mid P$, we have $b^2 \mid K$. In

$$(P - 2)k = K - X - 1$$

the right side is divisible by b^2, while $v_2(P - 2) = 1$. Because b is a power of two, $b^2/2 \mid k$, hence $b \mid k$. Reduction modulo three, using $3 \mid K, P, X$, gives $k \equiv 2 \pmod 3$. Since $b \equiv 1 \pmod 3$, $k/b \equiv 2 \pmod 3$, so $k = b(3k_0 + 2)$ for an integer k_0.

The recurrence also gives $u_{2j+1}(A) \equiv (-1)^j \pmod{A^2}$. As X is odd, $A^2 \mid C+1$, while $2b^2 \mid B_* + 1$. Thus $2b^2 \mid C - B_*$. The constructive A is a multiple of four, so $v_2(A-2) = 1$ and $h = (C-B_*)/(A-2)$

is divisible by b^2, hence by b. The normalizations in (6) therefore retain the actual constructive witnesses; we have not postulated arbitrary cofinality of Pell witnesses in chosen residue classes.

We have $\Delta > 0$, $S \mid T$, and b, Q positive squares. Lemma 3.2 supplies a positive z making a a square. Invoke the interpolation theorem to choose ξ divisible by this z. Write $q = T/S$, choose a root c of π modulo ξ, and request $y \equiv c - q \pmod{\xi}$. The theorem supplies such a y with d a square. Put $x = \xi/z$. Now $\Sigma \mid \Theta$, and the integral choice

$$m = \Theta/\Sigma + \sqrt{a} + \sqrt{d}$$

makes (12) vanish. This completes both directions.

4.6 Exact degree

All ten supplied variables have degree one; every fixed port has degree zero. The decisive register degrees are

Register	X	Y	A	C	K	Q
Degree	22,964	11,863	46,694	46,696	81,524	326,092
Register	Δ	a	d	Σ	Z	F_{10}
Degree	187,881	840,069	4,109,198	233,472	233,490	9,152,284

In the inner norm expression $\Sigma^2 d$ uniquely has largest degree, 4,576,142. In the outer expression $\Sigma^4 d^2$ dominates the subtracted $4\Sigma^4 ad$, whose degree is only 5,883,155. This proves the claimed exact degree once the leading form is nonzero. The independent checker uses the actual fixed payloads and computes a nonzero coefficient at the upper degree on the full diagonal, modulo three specified primes. Already one such nonzero residue proves that the integral leading coefficient is not zero; three are retained as independent transcription controls. This is an exact lower-bound certificate, not a random evaluation heuristic.

5 Nine positive variables: congruence-separated norms

5.1 Source and Jones registers

For the positive-coordinate form the supplied variables are

$$(\epsilon_+, g_+, h_+, s, w, \phi_+, i, j, n_+) \in \mathbb{Z}^9_{>0}.$$

Set $b = \epsilon_+ + 1$, $g = 2g_+$, $h = 2h_+$, $\phi = \phi_+ - 1$, $n = n_+ - 1$, and $B = a_c b^2$. Use the common registers (5), and put

$$\begin{aligned} L_c &= 2bB^{5741}, & R &= L_c^2 + (S_c + T_c)L_c + L_c - 1 - S_c, \\ N &= 2^{16012040} B^{2966}, & G_c &= 32bB^{183}. \end{aligned} \tag{14}$$

The calibrated block length L_c and the ratio threshold N are different. When $b = 2^s$, $B = 2^v$ with $v = \kappa + 2s$, the population calculation gives threshold exponent

$$5947v - 30s - 20 = \log_2(N^2).$$

This identity is the reason a polynomial square threshold is available without another variable. The source estimates prove $R, N \geq 8$ on all positive assignments, even before the margin imposes an upper bound on g. In fact J_c, S_c, T_c are positive and $R > L_c^2$.

The retained Jones ratio registers are

$$\begin{aligned}
&U = N^2 w, && Y = N^2 s, && M_0 = RY, && P = 2M_0^2 U,\\
&K = R + 1 + h(P-1), && A = M_0(U+1), && B_1 = 2R+1, && W_1 = bw,\\
&C = B_1 + W_1 + \phi, && J = 4A - 5, && Q = 3W_1 C - 2(W_1^2 - 1), &&\\
&D = (A^2-1)C^2 + 1, && E = iJDC^2, && F = (A^2-1)E^2 + 1, &&\\
&G = A + F(CD + 1 - A), && H = B_1 + jC, && I = (G^2-1)H^2 + 1. &&
\end{aligned} \tag{15}$$

Define

$$\begin{aligned}
&a_0 = (P^2-1)K^2 + 1, && u = JF, && v = QF + J(H - C),\\
&\mu = G_c\{K^2 - 4(C - KY)^2\} - 8gK^2, && M_d = Q + H - C. &&
\end{aligned} \tag{16}$$

The imported ratio/source theorem, including its constructive direction, equates source existence with the four conditions

$$a_0 \in \mathrm{Sq}(\mathbb{Z}), \quad DFI \in \mathrm{Sq}(\mathbb{Z}), \quad \mu > 0, \quad u \mid v \tag{17}$$

on these coordinates. The source change in Section 2 meets its quantitative hypotheses. For instance $\mu > 0$ gives $g < G_c/8 = 4bB$[183], exactly the bound used there. The retained `Ratio.lean` and `NineRatio.lean`, and their accompanying ordinary source proofs, specify this interface. This branch does not depend on Sun's 2026 ten-integer theorem.

5.2 Why the even restrictions preserve completeness

On every supplied assignment, B, N_0, L_c, g, S_c are even, so R is odd. The integer P is even and h is even, hence K is even. Thus

$$a_0 \equiv 1 \pmod 4. \tag{18}$$

The fixed square multiplier used below makes the other radicand zero modulo four. This gives noncollision without an extra multiplier involving h or a large separation padding term.

For completeness, canonical source words are even because every w_i is positive. The constructive ratio proof chooses $K = \psi_P(R+1)$, where $\psi_P(0) = 0, \psi_P(1) = 1$ and $\psi_P(n+2) = 2P\psi_P(n+1) - \psi_P(n)$. Reduction modulo two gives $\psi_P(n) \equiv n \pmod 2$. As $R+1$ is even, K is even. In $K = R+1+h(P-1)$ the coefficient $P-1$ is odd, so its positive h is already even. The constructive witnesses are therefore represented by positive g_+, h_+, as required.

5.3 A common finalizer with strict polynomial majorants

All Jones registers used here are positive; in particular $A \geq 33280$, $u, v > 0$, $H - C > 0$, and $0 < v/u \leq M_d$. Let λ be either 8 or 2048, and choose polynomial integer majorants satisfying

$$V_0 = PK > \sqrt{a_0}, \qquad V_1 > \lambda\sqrt{DFI}.$$

Set

$$\begin{aligned}
&a_1 = \lambda^2 DFI, && \Omega = u(2\mu - 1),\\
&E_0 = un + v - \Omega M_d, && \Theta = E_0 - \Omega(V_0 + V_1).
\end{aligned} \tag{19}$$

The equation is

$$\boxed{F_9 = \left(\Theta^2 - \Omega^2(a_0 + a_1)\right)^2 - 4\Omega^4 a_0 a_1 = 0.} \tag{20}$$

Proposition 5.1. *On the supplied positive domain, existence of n_+ satisfying (20) is equivalent to the four conditions (17).*

Proof. The integer $2\mu - 1$ is nonzero, so $\Omega \neq 0$ unconditionally. Equations (18) and $a_1 \equiv 0 \pmod 4$ give $a_0 \neq a_1$. Lemma 3.1 produces signed integer roots r_0, r_1 with $\Theta/\Omega = r_0 + r_1$. A square $a_1 = \lambda^2 DFI$ also makes DFI a square by the rational-square-root lemma. Since the integer majorants are strict, $|r_i| \leq V_i - 1$. Therefore

$$Z = M_d + V_0 + V_1 + r_0 + r_1 \geq M_d + 2 > 0.$$

The selected factor equation becomes

$$n + v/u = (2\mu - 1)Z.$$

Its left side is positive; hence $\mu > 0$. Its right side is integral; hence $u \mid v$. Conversely, under (17), take any signed roots and the same Z, and set $n = (2\mu - 1)Z - v/u$. This is an integer at least two because $v/u \leq M_d$. It is an allowed natural value, and $n_+ = n + 1$ is positive. Substitution makes one of the norm's factors zero. □

5.4 The cheaper variant

Put $e = iJC$. Since $D - 1 = (A^2 - 1)C^2$, one has $F = e^2D^2(D - 1) + 1$. Set

$$\lambda = 8, \qquad V_1 = e(8D^2 - 4D - 1)GH.$$

The exact identity

$$\{e(8D^2 - 4D - 1)\}^2 - 64DF = e^2(8D + 1) - 64D > 0 \tag{21}$$

holds because $e \geq 7$ and $D \geq 9$. Since $I \leq G^2H^2$, it gives the strict majorant required by Proposition 5.1. Its additional exact identity

$$V_1^2 - 64DFI = \{e^2(8D + 1) - 64D\}G^2H^2 + 64DF(H^2 - 1)$$

exposes the leading cancellation and supplies a division-free degree certificate. The resulting polynomial $F_{9,\mathrm{f}}$ has exact degree 6,692,108 and 604 positive-domain operations.

5.5 The lower-degree majorant

The following polynomial majorant is inherited from the incoming support-majorants package [3]. Our change is to remove the separation padding and its multiplier, using the unconditional congruence separation above. Write

$$\begin{aligned} e &= iJC, & L_m &= CD + 1 - A, & N_* &= 2G + L_m, \\ N_1 &= 2e^2D(D - 1)L_m + 3C, & N_2 &= 2e^2(D - 1)L_m, & & \\ N_3 &= 2e^2(L_m - C), & N_4 &= 2e^2C, & & \\ c_m &= 256i(e^2 + 2) + 2048. & & & & \end{aligned}$$

Define

$$\begin{aligned} T_m &= e\{(1024D^2 - 512D - 128)N_* - 64N_1 - 40N_2 - 28N_3 - 21N_4\} + c_m, \\ \lambda &= 2048, \qquad V_1 = T_mH. \end{aligned} \tag{22}$$

All four N_j are the displayed polynomials; no quotient computes them. For completeness the uniform inequality, rather than only an asymptotic expansion, is recalled here.

Put

$$t_n = 1024D^6 - 512D^5 - 128D^4 - 64D^3 - 40D^2 - 28D - 21,$$
$$d_n = 33792D^5 + 10560D^4 + 4928D^3 + 2464D^2 + 1176D + 441.$$

Direct expansion gives $t_n^2 - 1024^2 D^{11}(D-1) = d_n > 0$, and $t_n > 0$ for $D \geq 1$. The exact remainders $N_* = D^j N_j + r_j$ are

$$r_1 = 3 - A, \qquad r_2 = 3CD + 3 - A,$$
$$r_3 = 2e^2(A-1)D^2 + r_2, \qquad r_4 = 2e^2(1 - A - C)D^3 + 2e^2(A-1)D^2 + r_2.$$

Thus the uncorrected polynomial part of T_m equals

$$e\frac{t_n}{D^4}N_* + e\left(\frac{64r_1}{D} + \frac{40r_2}{D^2} + \frac{28r_3}{D^3} + \frac{21r_4}{D^4}\right).$$

For $A \geq 3$ and $C, i \geq 1$,

$$\frac{eA}{D} < \frac{9}{2}i, \quad \frac{eC}{D} < \frac{3}{2}i, \quad \frac{e(A+C-1)}{D} < 6i.$$

The possible negative remainder is less than $252ie^2 + (765/2)i$, whereas c_m exceeds this plus 2048. Writing $X_m = e^2D^3(D-1)$, the identities $DF = X_m + D$ and $G = (X_m/D)L_m + CD + 1$, together with $\sqrt{X_m + D} < \sqrt{X_m} + D/(2\sqrt{X_m})$, give

$$2048\sqrt{DF}\,G < 1024eD^2\sqrt{1 - 1/D}\,N_* + 2048v_m, \qquad v_m = \frac{CD+1}{2eD\sqrt{1-1/D}} < 1.$$

Combining the estimates proves $T_m > 2048\sqrt{DF}\,G > 0$ uniformly. As $I \leq G^2H^2$, this proves $V_1 > 2048\sqrt{DFI}$, including all assignments with arbitrarily large witness variables.

5.6 The second cancellation and exact degree

The decisive degrees in the calibrated nine-variable construction are

Register	D	e	G	H	P	K
Degree	139,324	69,663	719,588	22,967	81,527	81,528
Register	V_0	V_1	Ω	E_0	a_0	a_1
Degree	163,055	1,090,866	791,147	814,115	326,110	2,181,732

Let $\delta_0 = V_0^2 - a_0 = K^2 - 1$ and $\delta_1 = V_1^2 - a_1$. The majorant's exact defect identity gives

$$\deg \delta_0 = 163056, \qquad \deg \delta_1 = 1253160.$$

Set $B_0 = E_0^2 - 2E_0\Omega(V_0+V_1) + \Omega^2(\delta_0+\delta_1)$. The two-root norm admits the exact second-cancellation identity

$$F_9 = B_0^2 + 4\Omega^2 V_0 V_1 B_0 + 4\Omega^4(\delta_0 V_1^2 + a_0\delta_1). \tag{23}$$

The three displayed terms have degrees, respectively,

$$5,670,908, \qquad 5,671,669, \qquad 5,509,376.$$

The middle term is uniquely highest with a positive, nonzero diagonal leading coefficient. A simple degree walk through the primary norm would miss this cancellation and give an overestimate.

The supplied division-free certificate constructs $D^8\delta_1$ using the exact majorant identity, then constructs D^8B_0 and $D^{16}F_9$. It is a different polynomial from the primary equation: its exact degree is 7,900,853. Subtracting $16 \deg D = 2,229,184$ proves $\deg F_9 = 5,671,669$. On the diagonal of the nine input variables, with $a_c = 2^{2134940}$ and $c_c = 2^{16012040}$, the leading coefficients are

$$\mathrm{lc}(F_9) = 2^{507} a_c^{2835574} c_c^{348}, \qquad \mathrm{lc}(D^{16}F_9) = 2^{667} a_c^{3950070} c_c^{476}.$$

Their roles are separately labeled in the exported metadata. Cancellation by D^{16} occurs only in this proof of degree; neither circuit divides.

6 A degree-52 equation in fourteen positive variables

This secondary result combines the incoming quotient absorber, a positive auxiliary gap, and a stronger canonical auxiliary congruence. It is useful when degree matters more than reaching the smallest available number of variables. Its supplied coordinates are

$$(J, F, \alpha, z, f, i, T, s, w, \tau, \eta, \zeta, e, \gamma) \in \mathbb{Z}_{>0}^{14}.$$

Use the same four native fixed ports and define

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y,\\
H &= 4a + 3, & \Delta &= a^2 + H, & \ell &= EkY, & b_0 &= X + ac,\\
G &= 1 + \Delta c^2, & U &= 4q^4,
\end{aligned}$$

and

$$\begin{aligned}
R &= \{(q-1)(q-F) + \alpha\}(q^2-1) + (M_C + qM_F)J,\\
r &= k - R - 1, \qquad D_* = Ub_0 + H(U\gamma - r), \qquad M_* = D_*^2 - U^2 G,\\
I &= ic^2, \qquad A_{\mathrm{aux}} = \Delta(f^2-1), \qquad W = I^2 - A_{\mathrm{aux}},\\
V &= f^2(cT - R) - c, \qquad y = V + e,\\
N_a &= A_{\mathrm{aux}}(V^2 - y^2) + y^2,\\
N_0 &= \tau^2 - \ell(\ell + k), \qquad N_t = (K_0 + w)C + q - F - z(q-1).
\end{aligned}$$

The equation is

$$\boxed{P_{14} = M_*^2 + W^2 + N_a^2 + (N_0 N_t - 1)^2 - 1 = 0.} \tag{24}$$

6.1 A minimum block and soundness

For $f = 1$, $A_{\mathrm{aux}} = 0$ and $W = I^2 > 1$. For $f \geq 2$, $A_{\mathrm{aux}} > 1$ and N_a cannot vanish. Otherwise $A_{\mathrm{aux}}V^2 = (A_{\mathrm{aux}}-1)(V+e)^2$ would make $A_{\mathrm{aux}}(A_{\mathrm{aux}}-1)$ a rational square. Neither factor involving V can be zero because $e > 0$, and the consecutive product lies strictly between $(A_{\mathrm{aux}} - 1)^2$ and A_{aux}^2. Hence always $W^2 + N_a^2 \geq 1$. Equality forces $W = 0$ and $N_a = \pm 1$. With $A_{\mathrm{aux}} = I^2$, reducing $N_a = I^2V^2 - (I^2-1)y^2$ modulo four excludes the negative sign. Thus equation (24) is equivalent tuplewise to

$$M_* = 0, \quad W = 0, \quad N_a = 1, \quad N_0 N_t = 1.$$

Furthermore $N_0 = -1$ is impossible. Here $\ell = nk$ with $n = XY^2 \geq 2$, so it would give $\tau^2 - n(n+1)k^2 = -1$. For a solution of minimum positive k, multiplication by the unit

$2n+1-2\sqrt{n(n+1)}$ gives a new second coordinate $k' = (2n+1)k - 2\tau$ with $0 < k' < k$, a contradiction. The inequalities follow from $nk < \tau < (n+1/2)k$. Thus $N_0 = N_t = 1$.

Map a new zero to the incoming fourteen-variable construction by $T_{\mathrm{old}} = fT$ and $y_{\mathrm{old}} = |V + e|$. The identity

$$V = c(T_{\mathrm{old}} f - 1) - Rf^2$$

is exact, and $y_{\mathrm{old}} > 0$ follows from $N_a = 1$ and $I^2 > 1$. All old residuals vanish. The retained absorber proof then recovers its omitted positive quotient before using the strong Pell rank theorem; the order of that recovery is essential. For reference, it first derives $C > 0$, $q \geq 16$, and $3q < R < q^4 - 2$ from transport. The modulus $U/\gcd(U, H)$ exceeds $R+1$, because $\gcd(U, H) \leq 3$. First-norm classification supplies an early index bound; only then is the strong rank theorem applied to recover the original compiler witness. The complete inherited proof is retained in `fourteen/upstream/FOURTEEN_PROOF.md`.

6.2 The stronger auxiliary congruence

It remains to show that requiring T_{old} to be a multiple of f does not lose a canonical witness. At a normalized compiler point the actual packed index is $R \equiv 3 \pmod 4$. Put $A = a + 2$, so $\Delta = A^2 - 1$, and use the ordinary Pell sequences χ_A, ψ_A. The established completion chooses

$$c = \psi_A(R), \quad m_0 = 2cR, \quad f = \chi_A(m_0), \quad I_0 = \psi_A(m_0), \quad \rho = \Delta I_0,$$

and $V = \chi_\rho(R)/\rho$, $y = \psi_\rho(R)$. The strong coordinate is chosen so that $I = ic^2 = \rho$. In particular $\rho^2 = \Delta(f^2 - 1)$ and $c^2 \mid \rho$.

For $R = 2r_0 + 1$, the odd quotient polynomial Q_{r_0} satisfies

$$\chi_z(R) = zQ_{r_0}(z^2), \quad Q_{r_0}(1 - A^2) = (-1)^{r_0}\psi_A(R), \quad Q_{r_0}(0) = (-1)^{r_0}R.$$

Since r_0 is odd,

$$V = Q_{r_0}(\rho^2) \equiv -c \pmod{f^2}. \tag{25}$$

Also $V \equiv -R \pmod c$ and $f^2 \equiv 1 \pmod c$. Therefore $o = (V+c)/f^2$ is positive integral, $c \mid o+R$, and $T = (o+R)/c$ is positive integral. This is precisely the new formula $V = f^2(cT - R) - c$. The norm identity gives

$$(\rho^2 - 1)(y^2 - V^2) = V^2 - 1 > 0,$$

because $R \geq 3$, $\rho > 1$, and $V \geq 4\rho^2 - 3 > 1$. Hence $e = y - V > 0$. The retained quotient absorber then supplies its positive carrier γ. This constructs all fourteen positive coordinates.

6.3 Exact degree 52

The main residual has exact degree 26. Write $D_0 = b_0 + H\gamma$. Then

$$M_* = U^2(D_0^2 - G) - 2UHrD_0 + H^2r^2,$$
$$D_0^2 - G = 2ac(X + H\gamma) + (X + H\gamma)^2 - Hc^2 - 1.$$

Its leading form is $128w^2s^3(\eta + \zeta)\gamma(\beta J)^{19}$. The strong residual W has degree 22; the gap identity

$$N_a = V^2 - (A_{\mathrm{aux}} - 1)e(2V + e)$$

gives degree at most 23, and $N_0N_t - 1$ has degree at most 24. Only M_*^2 reaches degree 52. Thus

$$(P_{14})_{52} = 16384w^4s^6(\eta + \zeta)^2\gamma^2(\beta J)^{38} \neq 0.$$

The circuit has exactly 79 live operations. The stronger congruence (25) is already implicit in the canonical Pell completion; its use here pays for the extra gap coordinate without increasing the fourteen-variable evaluation cost.

7 Verification, dependencies, and formal status

7.1 What the archive checks

The top-level `verify.py` runs the finite checks offline, using Python's standard library. It rebuilds the literal circuits, checks their live-gate counts, and validates the retained fixed payloads. Individual reports give the exact check counts and hashes; the principal evidence is:

- Complete support isolation: all 49,632 interactions, 94 intended hits, and the five-collision legacy negative control. A second checker independently parses the residual expressions and reconstructs the emitted sparse polynomial, rather than trusting the builder's source grouping.
- Fixed-coefficient modular evaluation of the displayed ten-variable registers and their universal congruences, together with exact symbolic identities for the two-root extraction and complete degree metadata.
- Independent finite ring expansion of the majorant identities and the second-cancellation identity, plus actual-coefficient comparisons of the majorant certificate with D^{16} times the primary equation.
- Exact degree certificates using leading forms and nonzero coefficients at known upper degrees; these prove degree, rather than merely guessing it from a few evaluations.
- The fourteen-variable main-residual expansion, gap identity, coefficient binding, and full univariate degree replay modulo three primes, with targeted canonical Pell and minimum-block controls.
- Regeneration of the fixed native coefficient tuple from finite machine/tile data, with the inherited finite geometry and carry checks where supplied by its generator.

The finite checks establish finite identities, transcriptions, and certificate claims. They do not establish unsolvability by bounded search, and no sufficiently long simulation could do so. The unbounded existence equivalences are the mathematical proofs in this report and the explicitly retained inherited theorem chain.

7.2 Three different proof levels

1. **New ordinary arithmetic proofs.** The positive-word source transformation, the two-radical separation, the coprime guard, preservation of the new congruence restrictions, the nine-variable finalizer, and the fourteen-variable witness reconstruction are proved here. These use elementary integer arithmetic, classical Pell/Lucas theorems, and the stated source interfaces, all formalizable in ZFC.
2. **Inherited unbounded theorems.** The native coefficient compiler and K215 calibration identify the literal source with syntactic ZFC refutations. The nine-variable branch uses the Jones ratio theorem; the ten-integer branch additionally uses Sun's modified Lucas interpolation theorem from the 2026 preprint. Their use is neither a new proof of those results nor a claim of independent external peer review.
3. **Proof-assistant artifacts.** The pinned repository has a separate external Lean theorem for a closed generic nine-variable coefficient recipe. That theorem is not a kernel check of these new small literal circuits. Its status document also distinguishes the external equivalence from an exported object-language ZFC derivation of the two quoted sentences. This release does not supply a new end-to-end Lean/Rocq certificate or such an object-language derivation.

The native bundle retains the relevant exact source files, licenses and source pins, including the portable K215 endpoint source closure where available. Rebuilding that Lean endpoint requires its stated toolchain; it was not rerun in this session. The separate source-chain ledger records this boundary. The independent reviews in this archive are additional AI-assisted reviews within the session, not external human peer review.

7.3 Reproduction and integration

From an extracted archive, run

```
python3 verify.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first command is the finite arithmetic audit; TeX is needed only to rebuild the report. The exact per-component commands and output names are listed in `README.md`. The manifest records hashes for delivered files. Generated certificate circuits are clearly distinguished from the recommended primary circuits. No network access or search for a refutation is part of the polynomial generators.

For repository incorporation, retain this as a new incoming construction and update the frontier by domain and metric. Its natural first integration targets are the source extraction lemma, the universal residue identities, the two-root norm lemma, and the guard theorem. These are compact formal proof obligations. An end-to-end formal claim should be made only after connecting the literal emitted coefficient maps to the retained semantic theorems and exporting the desired object-language formulation if required.

8 What is new, and what remains open

The main improvement is a structural reduction in the number of radicands that must be combined. The ten-integer equation no longer carries the base square, an auxiliary Lucas square, and positivity as separate radical requirements. The coprime product $r = bQ$, together with residues modulo three, recovers both square conditions from one square guard. Residues modulo the even base distinguish this guard from the Lucas certificate. A norm with four factors therefore replaces the earlier eight-factor norm. This is responsible for most of the ten-variable degree reduction.

The nine-positive reduction exploits the same separation principle with an easier parity normalization. It can discard both the extra multiplicative separator and the large gap padding of the old majorant finalizer. The second defect expansion then exposes a further leading cancellation. The source transformation is complementary: it reduces the number of fixed source records, improving circuit costs for both branches without increasing the target span. It requires the repaired finite support certificate; merely reusing the earlier targets would be wrong.

The fourteen-variable result is a smaller local improvement. Its stronger auxiliary congruence and positive gap lower the degree of the residual that previously dominated, leaving the main norm as the unique degree-26 term before squaring. It does not compete with the few-variable constructions on variable count.

Several next directions are precise but unproved here. A shorter target code for the 94-record source could reduce both few-variable degrees. Further radicand combination would need a new arithmetic identity; the two-root lemma alone does not justify combining three radicals without separation. Reducing a supplied variable would require preserving constructive congruence freedom through the Lucas interpolation or finding a different source representation. Finally, a smaller

coherent native tuple would improve coefficient height, a metric untouched by the present word transformation. These are research targets, not claimed results.

A The complete positive quadratic source

The 31 coordinates, in the order used by the weight vector, are

$$(J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta).$$

Set $q = \beta J + 1$, $k = \eta + \zeta$, $C = q - F - \alpha$; these are affine abbreviations. The residual is left side minus right side in each row.

Row	Equation
1	$L = q^2 - 1$
2	$N = q(L + 1)$
3	$X = wq$
4	$a = E + sN$
5	$E = X(a - E)$
6	$c = k(a - E) + \eta$
7	$P = E(c - \eta) + k$
8	$\tau^2 = P(P - k) + 1$
9	$D = X + ac + \gamma(4a + 3)$
10	$\Delta = a^2 + 4a + 3$
11	$C_2 = c^2$
12	$D^2 = \Delta C_2 + 1$
13	$k = jc - V + 1 + hE$
14	$jc - V = (q - F)(N - q) - CL + (M_C + qM_F)J$
15	$(K_0 + w)C + (q - F) - z(q - 1) = 1$
16	$F_2 = f^2$
17	$V = of - c$
18	$g = y^2 - V^2$
19	$I = iC_2$
20	$B_{\mathrm{aux}} = I^2$
21	$B_{\mathrm{aux}} = \Delta(F_2 - 1)$
22	$y^2 - B_{\mathrm{aux}} g = 1$

The original sparse JSON has SHA-256

`1c336ba0b3c9827ffddcf7ce895cad9dca2907f907cde7484c7f387bb2cb1287.`

The plain expression parser and the sparse-record parser independently reconstruct the same polynomial data. The shifted transport convention of a different package is not substituted into row 15.

B Literal selective code and fixed coefficient recipe

The weights are

$$(w_1, \ldots, w_{31}) = (9, 25, 23, 139, 35, 151, 168, 21, 91, 181, 94, 52, 150, 180, 7, 135, \\ 38, 88, 183, 106, 178, 136, 20, 138, 8, 64, 105, 154, 37, 137, 120).$$

The corrected targets, tabulated by original source row, are

Row	1	2	3	4	5	6	7	8	9	10
Target	2792	86	2358	3929	5190	4238	1296	3634	451	3157
Row	11	12	13	14	15	16	17	18	19	20
Target	4384	4865	2124	685	1530	70	3290	4527	2696	2780
Row	21	22								
Target	1059	1874								

Together with the 22 residuals and the factorial formula (2), these tables specify every coefficient of $\mathcal{D}$. The sorted target order is retained in `code_positive.json`.

For completeness, the native tuple's finite numerical recipe is as follows. Let $\rho = 2^{121}$, $B_n = \rho^{8801}$, and

$$\mathcal{E} = \{0, \ldots, 942\} \cup \{983, \ldots, 992\} \cup \{1003, \ldots, 1032\} \\ \cup \{1063, 1064, 1095, 1096, 2195, 3296\}.$$

The retained literal 943 tiles have four ordered slot labels x_{it}. For $0 \le j < 44$ put $W_j = 2^{10+2j}$. At tile selector i, set

$$c_i = 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + W_{40} + W_{41} + W_{42} + W_{43}.$$

At copy exponents $o_t + j$, where $(o_0, o_1, o_2, o_3) = (1023, 1013, 1003, 983)$, put $c_{o_t+j} = W_{10t+j}$. At anchors $1095, 1096, 2195, 3296$ put, respectively, $W_{40}, W_{41}, W_{42}, W_{43}$. Set $c_{1063} = c_{1064} = 0$. Define

$$\mu_n = 1022 + \sum_{j=0}^{43} W_j,$$

$$Z_n = (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799},$$

$$D_C = \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2200} + \sum_{e \in \mathcal{E}} c_e(\rho^{4393-e} + \rho^{5495-e}),$$

$$D_R = \rho^{42},$$

$$M_C = B_n - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063},$$

$$M_F^{\mathrm{native}} = 4 + \mu_n(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1013+j} + \rho^{1065+j}) + \rho^{2195} + \rho^{3296} + Z_n.$$

The four ports are

$$\beta = B_n - 1, \qquad K_0 = D_C + B_n D_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{native}} + B_n - 1.$$

The tile list is regenerated by bounded arithmetic from the 430 literal actions of the retained K215 machine. The finite native generator and its source-chain ledger specify the exact ordering and all byte formats. None of these calculations requires knowing whether the machine halts.

References

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[7] J. P. Jones, *Universal Diophantine equation*, Journal of Symbolic Logic **47** (1982), 549–571. The exact ratio interface used here is also preserved in the pinned repository's `Ratio.lean` and `NineRatio.lean` sources.